

Sumit Srivastava

Researcher, Human-AI Dialogue & NLP

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Nationality: Dutch

Research Summary

Researcher in human-AI dialogue and natural language processing, grounded in psycholinguistics and cognitive science. PhD candidate at the University of Twente (defence 2026) and Marie Skłodowska-Curie fellow (NL4XAI). Work spans controlled language generation with LLMs, lexical alignment, and empirical user studies of trust, persuasion, and comprehension in human-AI interaction, with peer-reviewed publications at ACM IUI, ACM UMAP, and Computer Speech & Language.

Education

Ph.D., Computer Science (Human-Computer Interaction)

2020 – 2026

University of Twente

The Netherlands

- Dissertation: “The Other Wordly Effects: Lexical Alignment in Human-Agent Dialogues” (defence 2026).
- Promotor: Prof. dr. D. Heylen. Co-supervisors: Dr. M. Theune, Dr. A. Catala.

M.Sc., Artificial Intelligence (*Cum Laude*)

2019 – 2020

KU Leuven

Belgium

- Thesis: metaphor interpretation using Theory of Mind (Rational Speech Act model, ConceptNet).

M.Tech., Information Technology (CGPA 9.2/10)

2010 – 2012

IIT Allahabad

India

B.Tech., Computer Science

2006 – 2010

Uttar Pradesh Technical University

India

Publications

Journal Articles

Srivastava, S., Theune, M., Catala, A., & Reed, C. (under revision). *Your Word Against Mine: Lexical Alignment in Argumentative Dialogues with a Conversational Agent*. Cognitive Science.

Srivastava, S., Wentzel, S., Theune, M., & Catala, A. (2024). *Measuring and Implementing Lexical Alignment: A Systematic Literature Review*. Computer Speech & Language, 101731. Elsevier.

Conference Papers

Srivastava, S., Theune, M., & Catala, A. (2026, under review). *Weighted Decoding for Controlled Lexical Alignment in LLM-based Dialogues*. SIGDIAL.

Srivastava, S., Theune, M., Catala, A., & Reed, C. (2024). *Trust in a Human-Computer Collaborative Task With or Without Lexical Alignment*. UMAP Adjunct '24, ACM.

Srivastava, S., Theune, M., & Catala, A. (2023). *The Role of Lexical Alignment in Human Understanding of Explanations by Conversational Agents*. ACM IUI, 423–435.

Srivastava, S., & Tripathi, R. C. (2013). *Real-time Mono-vision Based Customizable Virtual Keyboard Using Finger Tip Speed Analysis*. HCI International 2013, Springer, 497–505.

Research Experience

Doctoral Researcher

2020 – Present

University of Twente, Human Media Interaction

The Netherlands

- Investigated how conversational agents can adapt word choice to improve user understanding, trust, and persuasion.
- Designed and ran controlled, between-subjects user studies; built Wizard-of-Oz and LLM-based conversational agents.
- Developed weighted-decoding methods for controllable lexical alignment with Llama, Phi, and BlenderBot.
- Supervised eight MSc/BSc students.

Marie Skłodowska-Curie Early-Stage Researcher

2020 – Present

NL4XAI (*EU Horizon 2020*)

- Funded fellow in the Interactive Natural Language Technology for Explainable AI network; research secondment at the Centre for Argumentation Technology, University of Dundee (Prof. C. Reed).

M.Sc. Researcher, Artificial Intelligence

2019 – 2020

KU Leuven

Belgium

- Automatic interpretation of metaphors using Bayesian inference (Rational Speech Act) and ConceptNet; proposed and developed the topic independently.

Fellowships & Awards

- Marie Skłodowska-Curie Fellowship, NL4XAI (EU-funded), 2020–present.
- M.Sc. Artificial Intelligence awarded *Cum Laude*, KU Leuven.

Industry & Engineering Experience

Senior Backend Engineer

2025 – Present

Tibo Energy

The Netherlands

- Industrial energy-management platform (TypeScript, Node.js, MySQL, Cassandra).

CTO (part-time / freelance)

2021 – 2025

BrainBite; ad:personam GmbH

NL / Remote

- Technical leadership of AI and platform development: LLM content generation and evaluation design (BrainBite); programmatic-advertising DSP (ad:personam).

Technical Lead & Senior Engineer

2013 – 2019

MiQ Digital

India

- Led a team of seven; built real-time ad-bidding systems at 1.5M requests/second.

Skills

Research: Experimental design, quantitative & qualitative analysis, statistics, scientific writing, user studies

NLP / AI: LLMs, controllable text generation, dialogue systems, psycholinguistics, RSA modelling

Programming: Python, Java; PyTorch, Keras, spaCy, NLTK, Rasa, DialogFlow

References

- Dr. Mariët Theune, Associate Professor, Human Media Interaction, University of Twente. m.theune@utwente.nl
- Dr. Alejandro Catala, Assistant Professor, CiTIUS, University of Santiago de Compostela. alejandro.catala@usc.es
- Prof. Chris Reed, Professor, Centre for Argumentation Technology, University of Dundee. chris@arg.tech